

NAV-1 & NAV-2

SUBTOPIC M-ST01A

SOURCE: Q1-Q5 BANK

PHYSICS TRUTH FIRST

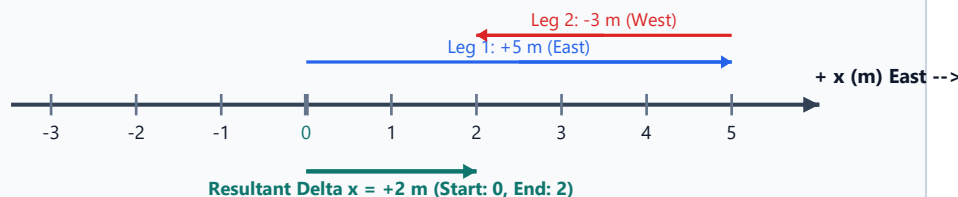
{{ LESSON_TITLE: Reference Frame, Coordinate Axis, Distance & Displacement }}

Core Pedagogy: SEE THE MOTION FIRST -> REALIZE THE VECTOR RELATION -> UNDERSTAND SIGN CONVENTIONS

{{ NAV_ROUTINE: Every Numerical Problem - First 20 Seconds }}

1. Draw a tiny coordinate line or functional sketch before touching equations.
2. Choose the positive direction (+ East / + Up) explicitly: sign errors stem from unchosen frames.
3. Translate hidden words into numerical physics data:
'Starts from rest' => $u = 0$ | 'Comes to a stop' => $v = 0$ | 'Highest point' => $v_{\text{top}} = 0$.
4. Model Gate (NAV-2): Is acceleration constant? (Use UVATS) | Is velocity constant? (Use $s = vt$)
5. Avoid formula roulette: Select the single relation that directly bypasses unneeded unknowns.

{{ VISUAL_PRIMITIVE: 1D Coordinate Line & Return-Path Vectors }}



{{ PHYSICAL_INVARIANT: Path Length vs Coordinate Difference }}

- * Distance (Scalar): Total cumulative path length travelled regardless of direction.
Distance is always positive and strictly monotonically non-decreasing over time: $d \geq 0$.
- * Displacement (Vector): Net change in position comparing only endpoints:
 $\Delta x = x_{\text{final}} - x_{\text{initial}}$. Displacement can be positive, negative, or zero.
- * The Reversal Trap: Whenever an object reverses direction, Distance > |Displacement|.
- * Magnitude Bound: $|\text{Displacement}| \leq \text{Distance}$. Equality holds ONLY for unidirectional straight motion.

PHYSICS WORKED EXAMPLE

{{ EXAMPLE: Two-Leg Reversal Journey }}

Problem: A student starts at origin $x = 0$, walks 6 m East, then turns and walks 2 m West.
Calculate: (a) Total Distance, (b) Final Position, (c) Net Displacement.

Physics Reasoning Story (Words --> Picture --> Facts --> Target --> Relation):

1. Define Frame: Let East be positive (+x direction). Origin is the school gate $x = 0$ m.
2. Leg 1 Vector: $\Delta x_1 = +6$ m (East). Position after Leg 1: $x_1 = 0 + 6 = +6$ m.
3. Leg 2 Vector: Moves West => $\Delta x_2 = -2$ m. Position after Leg 2: $x_2 = +6 + (-2) = +4$ m.
4. Calculate Distance: $d = |\Delta x_1| + |\Delta x_2| = 6 \text{ m} + 2 \text{ m} = 8 \text{ m}$. (Path length accumulation)
5. Calculate Displacement: $\Delta x = x_{\text{final}} - x_{\text{initial}} = (+4 \text{ m}) - (0 \text{ m}) = +4 \text{ m}$ (East).
6. Verification Check: $|\Delta x| = 4 \text{ m} < 8 \text{ m} = \text{Distance}$. Reversal detected and verified. [PASS]

GUIDED PRACTICE * TRY WITH ME

{{ GUIDED_PRACTICE_Q1 }}: Start at $x = -2$ m. Walk 5 m East, then 7 m West.

- * Step 1: Mark initial coordinate $x_0 = \underline{\hspace{1cm}}$ m.
- * Step 2: Displacement vector $\Delta x_1 = \underline{\hspace{1cm}}$ m, $\Delta x_2 = \underline{\hspace{1cm}}$ m.
- * Step 3: Final coordinate $x_{\text{final}} = x_0 + \Delta x_1 + \Delta x_2 = \underline{\hspace{1cm}}$ m.
- * Step 4: Total Distance $d = |\underline{\hspace{1cm}}| + |\underline{\hspace{1cm}}| = \underline{\hspace{1cm}}$ m.
- * Step 5: Net Displacement $\Delta x = x_{\text{final}} - x_0 = \underline{\hspace{1cm}}$ m.

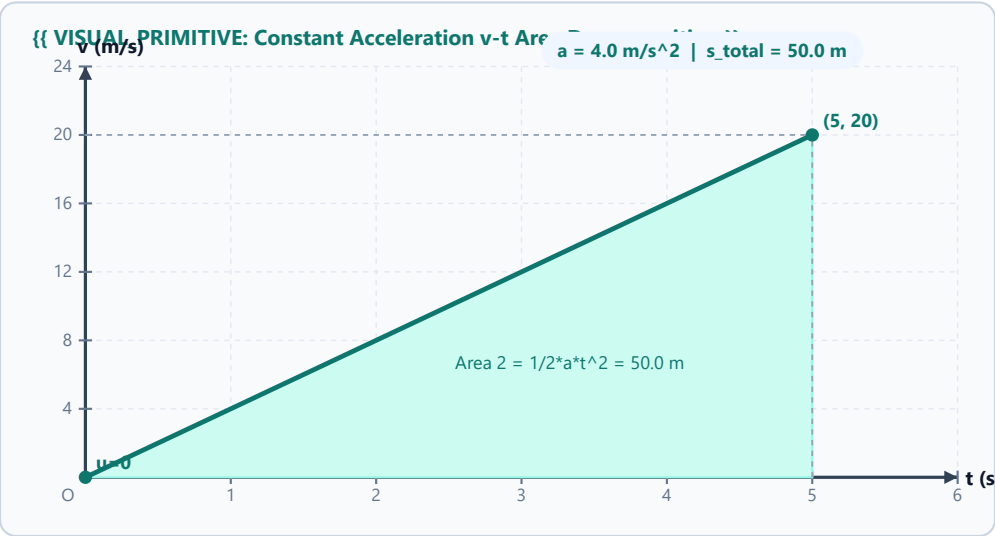
{{ GUIDED_PRACTICE_Q2 }}: Can an athlete complete one lap of a 400 m circular track

and have a displacement of 400 m? Explain using the definition of displacement.

Your physics explanation: _____

M-ST04A GRAPHS
 SLOPE & AREA DECODER
 SEE -> REALIZE -> UNDERSTAND

{{ LESSON_TITLE: Velocity-Time Area as Accumulated Movement & Kinematic Derivations }}



GRAPH MISCONCEPTION CLINIC

{{ MISCONCEPTION: Graph Height vs Graph Slope }}

* **Misconception 1: 'Higher point on x-t graph means higher speed.'**
 Correction: Height on an x-t graph indicates position (distance from origin).
 Speed is the SLOPE (rate of change). A flat horizontal line at x = 100 m means v = 0.

* **Misconception 2: 'Area below zero velocity axis is still positive distance.'**
 Correction: In v-t graphs, area below the v = 0 line represents NEGATIVE displacement (travel in the negative coordinate direction).
 To compute Total Distance: Add absolute areas |Area_above| + |Area_below|.
 To compute Net Displacement: Take signed sum: Area_above - Area_below.

* **Model Validity Guardrail: The kinematic equations ($s=ut + \frac{1}{2}at^2$, $v^2=u^2+2as$) are valid ONLY when acceleration is constant. For variable acceleration, area must be integrated.**

CORE 1 -> CORE 2 TRANSFER PIPELINE

{{ CORE2_TRANSFER_ROADMAP: Practice Set Progression }}

You have now mastered the conceptual representations of 1D Motion.
 In Core 2, you will encounter the audited PYQ & ExamSIDE transfer questions:

- * Assimilation Set 1: Averages, Speed vs Velocity, Harmonic Mean
- * Assimilation Set 2: Constant Acceleration Core (UVATS selection)
- * Assimilation Set 3: Polynomial motion & braking vehicle pairs
- * Assimilation Set 4: Free fall & vertical gravity stages
- * Assimilation Set 5: Relative motion & common acceleration cancellation
- * Assimilation Set 6: Velocity-Time & Position-Time graph interpretation

Transfer Rule: Always attempt before revealing H1-H3 hints. Check methods after finishing.

{{ DERIVATION_WALKTHROUGH: Why $s = ut + \frac{1}{2}at^2$ Works Physically }}

1. Geometry of the Velocity-Time Profile:
 Under constant acceleration 'a', velocity rises linearly from initial speed 'u' to 'v' in time 't'.
 The region under the graph decomposes into a base rectangle and an upper triangle.

2. The Base Rectangle Area (Displacement if speed never increased):
 $\text{Area}_{\text{rect}} = \text{base} \times \text{height} = t \times u = ut.$ (The distance covered at steady speed u)

3. The Upper Triangle Area (Extra displacement gained from acceleration):
 $\text{Area}_{\text{tri}} = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times t \times (v - u).$
 Since acceleration is defined as $a = \frac{(v - u)}{t} \Rightarrow (v - u) = at.$
 Substitute: $\text{Area}_{\text{tri}} = \frac{1}{2} \times t \times (at) = \frac{1}{2} at^2.$

4. Total Integrated Displacement: $s = \text{Area}_{\text{rect}} + \text{Area}_{\text{tri}} = ut + \frac{1}{2} at^2.$
 Physical Meaning: Initial movement + additional speed gain = total position shift.